

Theory of Spectral Relativity

Version 4: The Universal Attractor

Part I: Proven Foundations Part II: The Corrected
KPP Flow

Part III: The Gauge Condition Part IV: External
Validations

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Abstract

We present Version 4 of the Theory of Spectral Relativity, rebuilt around three logically separated blocks rather than a single derivation chain. **Block 1** establishes the topological kink $\Phi(x) = \tanh(\alpha x)$ as the unique finite-action minimizer of a double-well variational functional, from which its exact Pöschl–Teller stability spectrum and three pure invariants $J_1 = 6$, $J_2 = 6/5$, $J_3 = 1/\sqrt{3}$ follow. **Block 2** corrects a sign error identified in the coupled KPP flow formulation, present since the June 2026 source and re-

produced unchanged in Versions 2 and 3: the published stationary equation $D\rho'' + \lambda\rho - \gamma\rho^2 = 0$ does not admit the announced sech^2 profile as a fixed point; the corrected form $D\rho'' - \lambda\rho + \gamma\rho^2 = 0$ restores it exactly in the decoupled scalar case. Solved numerically without an imposed ansatz, the coupled system generically produces a *simple* zero ($n = 1$) of the net density $\rho_{\text{net}} = \rho_+ - \rho_-$, consistent with the generic case already anticipated by the Forbidden Multiplicities Theorem. **Block 3** establishes that selection of the special case $n = 2$ — required for the attractor $\Theta \rightarrow -1/2$ — comes from the gauge condition $H4'$ (common centering of the two kinks), an explicit geometric postulate whose derivation from Blocks 1–2 remains open, despite six documented attempts and a methodological lead identified in the pulse-interaction literature (Rademacher 2013, Ohta 2001). Two unadjusted external validations — string tension and deconfinement temperature in lattice QCD, Reynolds-independence of turbulence on the JHTDB database at three points ($\text{Re}_\lambda \approx 433, 610, 1257$) — remain separate from this internal chain and are unaffected by the correction. Complete verification scripts accompany each result.

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Revision Note

This version corrects a sign error in the coupled KPP flow formulation and clarifies the exact logical status of the attractor $\Theta \rightarrow -1/2$. A thorough audit — conducted jointly with several AI assistants (Claude, ChatGPT, Kimi), each verifying and correcting the others' results — established that this attractor is not an independent consequence of the KPP system, but a corollary of the gauge condition $H4'$, which remains an underived geometric postulate.

Nothing is silently deleted. Every Version 3 result that survives is reproduced as-is; every requalified result is flagged with its new exact justification; the sign error is documented with its correction and its historical origin, traced to the June 2026 source rather than the Version 3 rewrite.

Following the anti-circularity protocol already applied in previous versions, failed derivation attempts are documented as negative results rather than silently discarded.

Part I

Proven Foundations

1 The Kink as Fundamental Object

1.1 The Variational Functional

Let $\Phi : \mathbb{R} \rightarrow (-1, 1)$ be a smooth field with $\Phi(-\infty) = -1$, $\Phi(+\infty) = +1$. Define the action functional

$$\mathcal{A}[\Phi] = \int_{-\infty}^{\infty} \left[\frac{1}{2}(\Phi')^2 + \frac{\alpha^2}{2}(1 - \Phi^2)^2 \right] dx, \quad (1)$$

a standard double-well (ϕ^4 -type) functional with degenerate vacua $\Phi = \pm 1$. This is the fundamental object of this theory: rather than postulating an evolution equation and studying its fixed points, we postulate $\mathcal{A}$ and study its minimizers directly.

1.2 Uniqueness of the Kink

Theorem 1.1 (Kink Uniqueness). *The unique (up to translation) monotone, finite-action critical point of $\mathcal{A}$ is*

$$\Phi(x) = \tanh(\alpha(x - x_0)).$$

Proof. We give the proof in five steps, each depending only on the previous ones.

Step 1 (Euler-Lagrange equation). Stationarity of $\mathcal{A}$ gives

$$\Phi'' = 2\alpha^2\Phi(\Phi^2 - 1).$$

Step 2 (First integral). Multiplying by Φ' and integrating,

$$\frac{1}{2}(\Phi')^2 - \frac{\alpha^2}{2}(1 - \Phi^2)^2 = H,$$

for some constant H . This step uses only the Euler-Lagrange equation; it uses neither the boundary conditions nor the sign of Φ' .

Step 3 ($H = 0$ from boundary conditions). Since $\Phi \rightarrow \pm 1$ and finite action forces $\Phi' \rightarrow 0$ at both ends, evaluating the first integral at $x \rightarrow +\infty$ gives $H = 0$. This determination of H is logically prior to, and independent of, the Bogomolny factorization performed in Step 4.

Step 4 (Bogomolny equation). With $H = 0$,

$$\Phi' = \pm\alpha(1 - \Phi^2).$$

Monotonicity from -1 to $+1$ selects the $+$ sign.

Step 5 (Quadrature and Cauchy-Lipschitz). A separable first-order ODE:

$$\int \frac{d\Phi}{1 - \Phi^2} = \int \alpha dx \implies \Phi(x) = \tanh(\alpha(x - x_0)).$$

The right-hand side is Lipschitz in Φ on any compact subset of $(-1, 1)$, so the solution is unique by Cauchy-Lipschitz. $\square$

Remark 1.2. *The proof is structured so that Step 3 (boundary condition, $H = 0$) is established before the Bogomolny factorization of Step 4 that it feeds into. Reversing this order would make the argument circular. Script 9 (Appendix A) numerically verifies this non-circularity.*

1.3 Relation to the KPP System

The connection to the coupled KPP system, discussed in detail in Part II of this document, is as follows: writing

$$\rho(\omega) = \frac{A}{\alpha} \Phi'(\omega) = A(1 - \Phi(\omega)^2) = A \operatorname{sech}^2(\alpha\omega),$$

this density ρ coincides with the functional form of the scalar fixed point of the corrected KPP flow (Part II, Theorem 5.1) — once the sign correction is applied. Theorem 1.1 above remains independent of this correspondence: it establishes $\Phi = \tanh$ by direct quadrature, without reference to the KPP system.

2 The Pöschl–Teller Spectrum and the Pure Invariants

2.1 The Hessian of the Kink Action

Theorem 2.1 (Stability Spectrum). *The second variation of $\mathcal{A}$ about the kink $\Phi = \tanh(\alpha x)$ is the Schrödinger operator*

$$L = -\partial_x^2 + 4\alpha^2 - 6\alpha^2 \operatorname{sech}^2(\alpha x),$$

a Pöschl–Teller potential with discrete spectrum $\lambda_0 = 0$ (translation zero mode) and $\lambda_1 = 3\alpha^2$ (shape mode), plus a continuum $\lambda \geq 4\alpha^2$.

Proof. Linearizing the Euler-Lagrange equation about $\Phi = \tanh(\alpha x)$ with $\Phi = \tanh(\alpha x) + \eta$ gives, to first order,

$$-\eta'' + (4\alpha^2 - 6\alpha^2 \operatorname{sech}^2(\alpha x)) \eta = \lambda \eta,$$

the standard Pöschl–Teller operator of depth parameter $\ell(\ell + 1) = 6 \Rightarrow \ell = 2$ (in units of α^2), whose spectrum is $\lambda_n = 4\alpha^2 - \alpha^2(\ell - n)^2$ for $n = 0, 1, \dots, \lfloor \ell \rfloor$. $\square$

2.2 The Pure Invariants

Three dimensionless quantities follow from the kink shape alone:

$$J_1 = 6, \quad J_2 = \frac{6}{5}, \quad J_3 = \frac{1}{\sqrt{3}}, \quad \lambda_1 = \frac{1}{J_3^2} = 3\alpha^2.$$

Remark 2.2. *These invariants are properties of the kink Φ alone, independent of which physical system (Yang–Mills, Navier–Stokes) the kink represents. They are computed once, from the universal Φ , and exported to each domain via the projection operators (Part V).*

3 The Phase Coordinate and Hypotheses H1–H3

3.1 The Phase Coordinate σ

$$\sigma := \frac{\mu - \lambda}{\mu + \lambda} \in (-1, +1).$$

This coordinate organizes three physical regimes on a single axis: $\sigma < 0$ (dissipative, Navier–Stokes), $\sigma = 0$ (marginal, glass transition), $\sigma > 0$ (confining, Yang–Mills).

3.2 H1–H3

H1 (Coupled KPP, parameter bounds). The coupled system (Part II, Eq. 2.1–2.2) holds with strictly positive constants $D, \lambda, \mu, \gamma, \kappa > 0$ and $\kappa < \gamma$.

H2 (Critical dimension). $d_c = 4$; the mean-field exponent $\nu = 1/2$ holds without logarithmic corrections (Appendix B).

H3 (Confining phase). $\mu > \lambda$, equivalently $\sigma > 0$.

Part II

The Corrected KPP Flow

4 The Historical Sign Error: Diagnosis and Correction

4.1 The System as Written

Version 2, its source (Guiffra, June 2026, *Riemannian Spectral Geometry: Yang-Mills, Navier-Stokes and the KPP Flow*, osf.io/sjndw), and Version 3 state the coupled KPP system as:

$$\partial_\ell \rho_+ = D \partial_\omega^2 \rho_+ + \lambda \rho_+ - \gamma \rho_+^2 - \kappa \rho_+ \rho_-, \quad (2)$$

$$\partial_\ell \rho_- = D \partial_\omega^2 \rho_- + \mu \rho_- - \gamma \rho_-^2 - \kappa \rho_- \rho_+, \quad (3)$$

and claim (Version 2, Eq. 17; Version 3, Eq. 5) that the stationary fixed point of the scalar case ($\kappa = 0$) is

$$\rho^*(\omega) = \frac{3\lambda}{2\gamma} \operatorname{sech}^2 \left(\sqrt{\frac{\lambda}{4D}} \omega \right). \quad (4)$$

Proposition 4.1 (Inconsistency of the historical formulation). *Profile (4) does not satisfy the stationary equation associated with (2) (with $\kappa = 0$):*

$$D\rho'' + \lambda\rho - \gamma\rho^2 = 0.$$

Proof. Direct substitution gives a residual

$$D(\rho^*)'' + \lambda\rho^* - \gamma(\rho^*)^2 = \frac{6\lambda^2}{\gamma} \cdot \frac{\cosh(\sqrt{\lambda/D}\omega) - 2}{\left(\cosh(\sqrt{\lambda/D}\omega) + 1\right)^2} \neq 0,$$

not identically zero (it vanishes only at isolated points where $\cosh(\cdot) = 2$). Verified by independent symbolic computation on both the French and English versions of the text, as well as on the June 2026 source, which contains the same equation and the same announced profile — confirming that the inconsistency predates the Version 3 rewrite. $\square$

The linearization of (2) near $\rho = 0$ further gives $\rho'' = -(\lambda/D)\rho$, whose spatial exponents are purely imaginary ($r = \pm i\sqrt{\lambda/D}$): the origin is a *center* (oscillatory behavior), not a saddle point. No real unstable manifold exists from which to construct a localized homoclinic profile — explaining, independently of the algebraic test above, why (4) cannot be a stationary solution of (2).

4.2 The Correction

Theorem 4.2 (Corrected Fixed Point). *The system*

$$D\rho'' - \lambda\rho + \gamma\rho^2 = 0 \tag{5}$$

admits (4) as an exact stationary solution.

Proof. Direct substitution: the residual $D(\rho^*)'' - \lambda\rho^* + \gamma(\rho^*)^2$ vanishes identically (symbolic computation, Script 10, Appendix A).

□

The correction simultaneously repairs the underlying dynamical structure: linearizing (5) near $\rho = 0$ gives $\rho'' = +(\lambda/D)\rho$, real exponents $r = \pm\sqrt{\lambda/D}$ — a saddle point, with a real unstable manifold, compatible with the homoclinic orbit that the sech^2 profile represents.

Remark 4.3. *This correction concerns only the sign of the reaction terms; it modifies neither the amplitude $A = 3\lambda/(2\gamma)$, nor the rate $\alpha = \sqrt{\lambda/4D}$, nor the sech^2 functional form of the profile announced in previous versions.*

The corrected coupled system reads:

$$D\rho_+'' - \lambda\rho_+ + \gamma\rho_+^2 - \kappa\rho_+\rho_- = 0, \quad (6)$$

$$D\rho_-'' - \mu\rho_- + \gamma\rho_-^2 - \kappa\rho_+\rho_- = 0. \quad (7)$$

5 The Decoupled System: Exact Scalar Fixed Point

For $\kappa = 0$, (6)–(7) reduce to two independent equations, each admitting the exact fixed point (4) with its own parameters:

$$\rho_+^{(0)}(\omega) = \frac{3\lambda}{2\gamma} \operatorname{sech}^2(\alpha_+\omega), \quad \rho_-^{(0)}(\omega) = \frac{3\mu}{2\gamma} \operatorname{sech}^2(\alpha_-\omega),$$

with $\alpha_+ = \sqrt{\lambda/4D}$, $\alpha_- = \sqrt{\mu/4D}$.

Crucial point: under H3 ($\mu > \lambda$), the amplitudes $A_+ = 3\lambda/2\gamma$ and $A_- = 3\mu/2\gamma$ are *rigidly determined* by λ and μ respectively — they are not free parameters, unlike the variational kink construction (Part I), where $A_\pm$ are adjustable construction parameters.

6 The Coupled System: Generic Selection of $n = 1$

6.1 The Order of the Zero of ρ_{net}

Proposition 6.1 (Simple zero of the decoupled system). *For $\kappa = 0$ and $\mu \neq \lambda$ (H3), the function $\rho_{\text{net}}^{(0)}(\omega) = \rho_-^{(0)}(\omega) - \rho_+^{(0)}(\omega)$ has, at its first zero crossing $\omega_0 > 0$, a zero of exactly order $n = 1$:*

$$\rho_{\text{net}}^{(0)}(\omega_0) = 0, \quad (\rho_{\text{net}}^{(0)})'(\omega_0) \neq 0.$$

Proof (numerical verification and genericity argument). For $D = \gamma = 1$, $\lambda = 1$, scanning $r = \mu/\lambda \in [1.001, 100]$, the crossing ω_0 exists and $(\rho_{\text{net}}^{(0)})'(\omega_0)$ does not vanish at any tested point (Script 11, Appendix A; values from -0.0003 to -13.2 across the tested range). By rescaling ($D = 1$, $\gamma = 1$, $\omega \rightarrow \omega/\sqrt{\lambda}$), only the ratio $r = \mu/\lambda > 1$ determines the transversality of the crossing; it degenerates continuously to zero only at the excluded limit $r \rightarrow 1$ (which violates H3). The codimension argument (implicit function theorem applied to $F(\omega; \text{param.}) = \rho_{\text{net}}^{(0)}$) indicates that $n = 2$ would correspond to a codimension-1 condition in parameter space — a measure-zero set, not the generic case. $\square$

Remark 6.2. *This proof establishes strong numerical evidence of genericity over the tested range; it does not constitute a complete analytical proof for all $(D, \lambda, \mu, \gamma)$. This reservation is explicitly maintained.*

6.2 Verification on the Real Coupled System ($\kappa \neq 0$)

The above result concerns the decoupled case. The decisive question is whether coupling $\kappa \neq 0$ modifies this conclusion.

Proposition 6.3 (Simple zero of the coupled system, direct numerical verification). *For $D = 1$, $\lambda = 1$, $\mu \in \{1.5; 2.0\}$, $\gamma = 1$,*

and $\kappa \in \{0.05; 0.1; 0.2\}$ (respecting $H1$, $\kappa < \gamma$), direct numerical resolution of the stationary system (6)–(7) via boundary-value problem (collocation method, `solve_bvp`, no imposed ansatz, initialized from the decoupled solution then followed by continuation in κ) converges to a non-trivial fixed point whose net density exhibits a simple zero:

$$\rho_{\text{net}}(\omega_0) = 0, \quad \rho'_{\text{net}}(\omega_0) \neq 0,$$

with $|\rho'_{\text{net}}(\omega_0)|$ increasing with κ (from 0.149 at $\kappa = 0$ up to 0.532 at $\kappa = 0.2$) — reinforcing, not weakening, transversality.

Proof (numerical, Script 12, Appendix A). See Appendix A, Script 12, for the complete code and detailed numerical output (table of ω_0 , $\rho'_{\text{net}}(\omega_0)$ values for each tested parameter set). Result independently reproduced by two distinct implementations, with numerical agreement to the decimal place. $\square$

Remark 6.4. *Numerical continuation fails beyond $\kappa \approx 0.2$ (solver non-convergence, not a physical signal of transition). Complete characterization of the coupled fixed point for κ close to γ remains an open problem.*

7 What the KPP Does Not Provide

Corollary 7.1 (Insufficiency of the KPP alone). *The corrected coupled KPP system (6)–(7), under hypotheses H1–H3 alone, does not select the double zero ($n = 2$) required for the attractor $\Theta \rightarrow -1/2$ (Part III). It produces, in all tested regimes, the generic case $n = 1$ of the Forbidden Multiplicities Theorem (Part III, Theorem 9.2(i)).*

This result is not an isolated weakness: it is consistent with the fact that the natural amplitudes of the coupled system, $A_{\pm} = 3\lambda_{\pm}/(2\gamma)$, are fixed by λ and μ separately and necessarily differ under H3. The equality $A_+ = A_-$ — required for $n = 2$ — is neither forbidden nor produced by the KPP dynamics: it simply is not its natural output.

Part III

The Gauge Condition $H4'$

8 $H4'$ as an Explicit Geometric Postulate

Definition 8.1 (Gauge condition $H4'$). *The spectral horizon ω_c coincides with the common center of the two fronts: writing Φ_+, Φ_- for the kink profiles governing ρ_+, ρ_- ,*

$$\Phi_+(\omega_c) = \Phi_-(\omega_c) = 0.$$

Unlike the presentation of previous versions, $H4'$ should not be read as directly forcing $A_+ = A_-$ from common centering alone. The exact mechanism, established below, distinguishes three logically independent ingredients.

9 The Double-Zero Mechanism: Three Separate Ingredients

Let $\xi = \omega - \omega_c$, and $\rho_{\pm}(\xi) = A_{\pm}(1 - \Phi_{\pm}(\xi)^2)$ with $\Phi_{\pm}(\xi) = \tanh(\alpha_{\pm}(\xi - \delta_{\pm}))$, where $\delta = \delta_+ - \delta_-$ measures the offset between the individual centers of the two kinks.

Proposition 9.1 (Decomposition of the zero). *The Taylor expansion of $\rho_{\text{net}}(\xi) = \rho_{-}(\xi) - \rho_{+}(\xi)$ about $\xi = 0$ decomposes as follows:*

- (i) **Spatial parity.** *If $\delta = 0$ (shared centers), the coefficient of order ξ^1 vanishes automatically, regardless of the relation between A_{+} and A_{-} — a consequence of the parity of $\rho_{\pm}$ in ξ (since $\Phi_{\pm}$ is odd, hence $\Phi_{\pm}^2$ is even).*
- (ii) **Zero-level cancellation.** *The coefficient of order ξ^0 equals $A_{-} - A_{+}$. Its cancellation — the requirement that ω_c actually be a zero of ρ_{net} , distinct from the centering condition (i) — forces $A_{+} = A_{-} =: A$.*
- (iii) **Non-degeneracy.** *With $A_{+} = A_{-} = A$ and $\delta = 0$, the coefficient of order ξ^2 equals $A(\mu - \lambda)/(4D)$, nonzero under $H3$ ($\mu \neq \lambda$).*

Proof. Direct symbolic computation (Script 13, Appendix A), verified independently on the parametrization $\alpha_{\pm} = \sqrt{\lambda_{\pm}/4D}$ with generic centers ω_{+}, ω_{-} , then specialized to $\delta = 0$. $\square$

Remark 9.2 (Correction of the Forbidden Multiplicities Theorem).

Version 3 stated (Theorem 6.1(ii)): “the only accessible orders are $n = 1$ (generic, $A_{+} \neq A_{-}$) and $n = 2$ (special, $A_{+} = A_{-}$).” This formulation omits a necessary condition: $n = 2$ requires $A_{+} = A_{-}$ and $\delta = 0$ (common centering) — not $A_{+} = A_{-}$ alone. With

$\delta \neq 0$, even $A_+ = A_-$ does not produce a double zero: the order- ξ^1 coefficient generically becomes nonzero again (proportional to $\tanh(\delta\sqrt{\mu/D})$, Script 13).

10 The $-1/2$ Theorem, Reformulated

Theorem 10.1 (Universal Attractor, corrected form). *Under H1–H3, assuming the two stationary profiles share a common center ($\delta = 0$) and that ω_c is defined by $\rho_{\text{net}}(\omega_c) = 0$, then:*

- (i) *center symmetry cancels the order-1 term;*
- (ii) *cancellation at the central point forces $A_+ = A_-$;*
- (iii) *H3 ensures the quadratic coefficient is nonzero.*

Consequently $\rho_{\text{net}}(\omega) = C(\omega - \omega_c)^2 + O((\omega - \omega_c)^3)$, $C \neq 0$, and

$$\Theta(\omega) \xrightarrow{\omega \rightarrow \omega_c} -\frac{1}{2}.$$

Proof. The last implication ($n = 2 \Rightarrow \Theta \rightarrow -1/2$) is a general analytic identity, independent of RST: if $f(\xi) = C\xi^m(1 + O(\xi))$, then $P = f'/f = m/\xi + O(1)$ (residue of the logarithmic derivative at a zero of order m , a classical fact of complex analysis), hence $\Theta = P'/P^2 \rightarrow -1/m$. For $m = 2$, $\Theta \rightarrow -1/2$. $\square$

Remark 10.2 (Explicit separation of mathematical content). *The relation $\Theta \rightarrow -1/m$ for a zero of order m is a textbook fact (residue*

of the logarithmic derivative, found in any classical complex analysis treatise). It depends on RST in no way. What RST establishes — and what is not a generic fact — is the precise mechanism (Section 2 above) that fixes $m = 2$ under $H4'$, from the coupled KPP equations. It is this mechanism, not the identity $\Theta = -1/m$ itself, that constitutes the theory's substantial contribution.

Proposition 10.3 (Insufficiency of the KPP alone to establish the prerequisite $\delta = 0$, $A_+ = A_-$). *The condition $\delta = 0$ used above does not result from the corrected KPP system (6)–(7) under $H1$ – $H3$ alone (Part II, Section 4): profiles derived from the coupled KPP system, solved numerically without imposing $\delta = 0$ or $A_+ = A_-$, generically produce $n = 1$. The condition $H4'$ must therefore be imposed as an additional hypothesis, external to the KPP dynamics.*

11 Attempts to Derive $H4'$: Documented Negative Results

Following the project's anti-circularity protocol, every attempt to derive $H4'$ from Blocks 1–2 is documented, whether it succeeds or fails.

(N1) Isoenergy, topological charge, coupled functional, Φ -coordinates (Version 2, methodological notes). None of these four routes derives $H4'$.

- (N2) **Mono-component energy matching** (Section 7.2, Version 3, retained). The equivalence $A_+ = A_- \Leftrightarrow E_+[\rho_+^*] = E_-[\rho_-^*]$ fails for the mono-component functional: $E_\pm \propto A_\pm^2/\alpha_\pm$, incompatible with $A_+ = A_-$ under H3 ($\alpha_+ \neq \alpha_-$).
- (N3) **Geodesic completeness of the spectral manifold** (Part V, Section 3). The spectral horizon there is defined from an individual positive metric density $\rho > 0$, not from the signed zero of ρ_{net} . The link confinement $\Rightarrow$ horizon $\Rightarrow n \geq 2$ does not pass through the KPP block itself but through the physical projections P_i , defined ad hoc.
- (N4) **Forbidden Multiplicities Theorem as a derivation.** Its proof already presupposes $H4'$ (via the Phase Transfer Lemma). It cannot therefore be used to derive $H4'$ without circularity.
- (N5) **Projective curvature.** Projective flatness $K = 0$ (Remark 6.3, Version 3) is a property of the solution ρ_{net} at the crossing point, not a property of the Riemannian metric (M, g) itself. No standard Riemannian geometry theorem imposes $K = 0$.
- (N6) **Berry connection on the Pöschl–Teller spectrum.** The levels $\lambda_0 = 0$, $\lambda_1 = 3\alpha^2$ never cross as α varies ($\lambda_1 - \lambda_0 = 3\alpha^2 \neq 0$): no exploitable spectral degeneracy.

Proposition 11.1 (Geometric overlap: partial result). *Let $\mathcal{G}(\delta) = \int_{-\infty}^{\infty} \rho_+^{(0)}(\omega) \rho_-^{(0)}(\omega - \delta) d\omega$ be the overlap integral of the decoupled*

profiles. $\mathcal{G}$ is maximized exactly at $\delta = 0$ (a consequence of the type of the Riesz rearrangement inequality for positive, even, unimodal functions), with $\mathcal{G}''(0) < 0$ (verified numerically, Script 14).

Remark 11.2. *This mathematical result is correct, but its physical interpretation remains open: the coupling term in (6)–(7) is $-\kappa\rho_+\rho_-$ (competitive), which would naively suggest, on an energetic reading, that the system tends to minimize rather than maximize overlap — the opposite of $H4'$. This reading cannot, however, be confirmed without a rigorous Lyapunov functional for the coupled system, which has not been established (a direct construction attempt fails: see Script 15). $\mathcal{G}(\delta)$ is proposed as a candidate geometric building block for a future derivation program, not as an accomplished derivation.*

12 Open Program: Pulse Interaction

The literature on pulse interaction in coupled reaction-diffusion systems offers an established methodological framework, potentially applicable to deriving $H4'$:

- **Rademacher (2013)**, *First and second order semi-strong interaction in reaction-diffusion systems*, SIAM J. Appl. Dyn. Syst.: establishes that first-order pulse interaction is *gradient-like* (in the sense of a Lyapunov functional) under verifiable

conditions, for a class of equations including Gray-Scott and Schnakenberg.

- **Ohta (2001)**, *Pulse dynamics in a reaction-diffusion system*, Physica D: derives an N -body-type equation of motion for a pair of interacting pulses, with attraction/repulsion regimes depending on parameters.

Open Question 12.1 (Adaptation to the case $D_+ = D_-$). *Rademacher's (2013) explicit framework relies on a diffusion-scale separation ($\varepsilon \ll 1$, $D_u \neq D_v$). The RST coupled system (6)–(7) has $D_+ = D_- = D$: no scale separation, direct rather than long-range-mediated coupling. Rademacher's explicit formulas therefore do not apply directly; only the method (matched asymptotic expansion, multiple scales) is potentially transferable. Adapting this method to the case $D_+ = D_-$, and determining the effective potential $W(\delta)$ selecting — or not — the inter-pulse distance $\delta = 0$, constitutes an open research program. Nothing guarantees a priori that this effective potential, once constructed, selects $\delta = 0$ rather than $\delta \rightarrow \infty$.*

Remark 12.2 (Final status of $H4'$). *$H4'$ is the central geometric postulate of this theory, functionally analogous (not in content) to the postulate of the constancy of the speed of light in special relativity: an underived physical principle whose quantitative consequences are verifiable. Its derivation — or the rigorous demonstration of its logical independence — constitutes the central open*

mathematical problem of the theory.

Part IV

The Riemannian Spectral Manifold

This part reproduces Section 8 of Version 3 in full; nothing here depends on the choice between the KPP-flow starting point and the kink starting point, since both share the same fixed-point density ρ^* (Part II, Eq. 4.2, now in corrected form).

13 Definition

Definition 13.1 (Spectral manifold). *Let $\rho : I \rightarrow \mathbb{R}_{>0}$ be a smooth spectral density on $I \subseteq \mathbb{R}$. The Riemannian spectral manifold is (I, g) with*

$$g_{\omega\omega}(\omega) = \frac{1}{\rho(\omega)}, \quad ds^2 = \frac{d\omega^2}{\rho(\omega)}.$$

The geodesic distance from ω_0 to ω_c is $s(\omega_0, \omega_c) = \int_{\omega_0}^{\omega_c} \frac{d\omega}{\sqrt{\rho(\omega)}}$.

Regions where ρ is large correspond to short metric distances (dense spectrum); regions where ρ is small correspond to large distances (spectral desert). A zero of ρ may create a point at infinite distance — a *spectral horizon*.

14 Logarithmic Geometry

Proposition 14.1. *The spectral metric $ds^2 = d\omega^2/\omega^2$ is flat in the coordinate $x = \log(\omega/\omega_c)$: $ds^2 = dx^2$. The manifold is isometric to $(\mathbb{R}, dx^2)$.*

Proof. $\omega = \omega_c e^x \Rightarrow d\omega = \omega dx \Rightarrow ds^2 = \omega^2 dx^2 / \omega^2 = dx^2$. $\square$

Proposition 14.2. *$(M, ds^2 = d\omega^2/\omega^2)$ is isometric to the vertical geodesic $u = 0$, $v = \omega > 0$ of the Poincaré half-plane $\mathbb{H}^2 = \{(u, v) : v > 0\}$ with metric $ds^2 = (du^2 + dv^2)/v^2$.*

Proof. On $u = 0$, $v = \omega > 0$: $ds^2|_{u=0} = dv^2/v^2 = d\omega^2/\omega^2$. $\square$

Remark 14.3. *This establishes that (M, ds^2) is isometric to a geodesic of $\mathbb{H}^2$, not that $M = \mathbb{H}^2$. Multiple manifolds contain geodesics isometric to $\mathbb{R}$; $\mathbb{H}^2$ is the natural minimal extension compatible with the one-dimensional logarithmic geometry, motivating but not resolving the open classification question of Part V.*

15 The Spectral Horizon

Proposition 15.1 (Horizon criterion). *If $\rho(\omega) \sim C(\omega - \omega_c)^\theta$ near ω_c : $\theta < 2$ gives $s(\omega_0, \omega_c) < \infty$ (accessible, no horizon); $\theta \geq 2$ gives $s = +\infty$ (spectral horizon).*

Proof. Near ω_c , $ds = d\omega/\sqrt{C(\omega - \omega_c)^\theta}$; $\int^{\omega_c} (\omega - \omega_c)^{-\theta/2} d\omega$ converges if and only if $\theta/2 < 1$. $\square$

Remark 15.2 (Clarification added in Version 4). *This criterion concerns the individual, positive-by-definition metric density $\rho > 0$ of the spectral manifold (Section 1 above), not directly the signed net density $\rho_{\text{net}} = \rho_- - \rho_+$ of Part III. Applying the horizon criterion to Yang-Mills confinement (Section 5 below) passes through a physical projection P_i relating these two objects; this relation is not automatic and constitutes one of the limitations identified in the $H4'$ derivation attempts (Part III, attempt N3).*

16 The KPP Fixed Point and its Curvature

With ρ^* as in Part II (Eq. 4.2, corrected form), the scalar curvature of $g^* = 1/\rho^*$ is

$$R^*(0) = -\frac{\lambda}{2D} < 0 \quad (\text{hyperbolic at center}), \quad R^*(\omega \rightarrow \infty) = +\frac{\lambda}{2D} > 0 \quad (\text{spherical at infinity})$$

The fixed-point manifold interpolates between hyperbolic and spherical geometry.

17 The Confinement-Regularity Duality

The Fundamental Duality. Let (M_ρ, g) be the Riemannian spectral manifold with $ds^2 = d\omega^2/\rho$. Then:

Yang-Mills: $\rho(\omega_c) = 0, \theta \geq 2 \iff s = +\infty \iff$ Confinement

Navier-Stokes: $E(k, t) > 0 \forall k \iff s < \infty \iff$ Global Regularity

Confinement = existence of a spectral horizon. Regularity = absence of a spectral horizon.

Part V

Physical Realizations

This part reproduces Section 9 of Version 3, with the precision of Section 4 above (remark on the role of projections P_i) now made explicit.

18 Yang-Mills: Closed-Form Propagator

In the effective Yang-Mills $SU(N)$ framework, $\gamma^2 = 11N/3$ and $\omega_c = \Lambda_{\text{QCD}}$. With $\rho_{\text{KL}}(\mu^2) = C_{\text{YM}}(\mu^2)^2 e^{-\mu^2/\Lambda^2}$, $C_{\text{YM}} = (\mu -$

$$\lambda)^2/(8\gamma D_{\text{KPP}}),$$

$$D_{\text{YM}}(k^2) = C_{\text{YM}}\Lambda^4 - C_{\text{YM}}\Lambda^2 k^2 + k^2 (2C_{\text{YM}} \log \Lambda - C_{\text{YM}} \log k^2) + O(k^3),$$

derived from KPP parameters alone (Appendix C gives the complete computation), with $D_{\text{YM}}(0) = C_{\text{YM}}\Lambda^4$ finite and nonzero, consistent with lattice data $D(0) \approx 8 \text{ GeV}^{-2}$.

19 Navier-Stokes: Spectral Flux and Two-Front Structure

In the four-variable NS system, $\rho_{\text{NS}}(k) = E(k)$, with KPP fixed point $E^*(k) = \frac{3\lambda}{2\gamma} \text{sech}^2(\alpha_0(k - k_0))$, $\alpha_0 = \sqrt{\lambda/4D}$. The spectral energy flux takes the two-front form:

$$\frac{\Pi(k)}{\varepsilon} = \frac{1 + \tanh(\alpha_{\text{inj}} \log(k/k_0))}{2} - \frac{1 + \tanh(\alpha_{\text{diss}} \log(k/k_\eta))}{2},$$

producing the $0 \rightarrow \varepsilon \rightarrow 0$ topology across the inertial range, with $\Pi(k_0)/\varepsilon = 1/2$ parameter-free.

Remark 19.1 (Distinction: $E(k)$ versus $\Pi(k)$). $E(k) = \rho_{\text{NS}}(k)$ is the KPP density itself; $\Pi(k) = \int_0^k 2\nu k'^2 E(k') dk'$ carries the additional dissipation weight $2\nu k'^2$, modifying the shape and width of the sigmoid relative to $E(k)$. This distinction governs the projection-operator discussion below.

Universal coupling formula. $\gamma^2 = 11N/3$ (Yang-Mills) and $C_B = 15/2$ (Navier-Stokes Batchov invariant) share

$$\kappa = \frac{d_{\text{sym}}(d_{\text{sym}} + 2)}{d_{\text{sym}} - 1} :$$

exact for $SO(3)$ ($d_{\text{sym}} = 3$, NS); a 3.9% finite- N discrepancy for $SU(3)$ ($d_{\text{sym}} = 8$, YM).

Part VI

External Validations

This part reproduces and extends the external validation results of Version 3. These results are independent of the KPP sign correction (Part II) and the requalification of $H4'$ (Part III): they compare a prediction derived from the theory to independently measured data, without readjustment.

20 Yang-Mills: Lattice QCD

In the effective Yang-Mills $SU(N)$ framework, $\gamma^2 = 11N/3$ and $\omega_c \equiv \Lambda_{\text{QCD}}$. From the single input parameter $\omega_c = 0.200$ GeV, $N = 3$:

$$\sigma_{\text{string}} = \frac{48\omega_c^2}{\pi^2} = 0.194 \text{ GeV}^2 \quad (\text{lattice: } 0.194 \text{ GeV}^2, \text{ deviation } < 0.5\%),$$

$$T_c = \omega_c \sqrt{\frac{N^2 - 1}{4.347N}} = 156.7 \text{ MeV} \quad (\text{lattice: } 155 \pm 5 \text{ MeV, deviation } 1.1\%)$$

$$T_c/\sqrt{\sigma} = 0.355 \quad (\text{lattice: } 0.356, \text{ deviation } 0.3\%).$$

Glueball mass ratios (comparison with Morningstar–Peardon lattice

data):

$$\begin{aligned}
M_{2++}/M_{0++} &= 1.348 \quad (\text{lattice: } 1.39 \pm 0.04, \text{ deviation } 3.0\%), \\
M_{0-+}/M_{0++} &= 1.510 \quad (\text{lattice: } 1.50 \pm 0.04, \text{ deviation } 0.7\%), \\
M_{0^{*}++}/M_{0++} &= 1.510 \quad (\text{lattice: } 1.54 \pm 0.11, \text{ deviation } 1.9\%).
\end{aligned}$$

Proposition 20.1 (Untested prediction). $M_{4++} = 2.787\text{--}2.805 \text{ GeV}$, currently unmeasured by lattice QCD. Testable at PANDA/BES-III.

21 Navier-Stokes: JHTDB Validation, Three Reynolds Points

Dataset	Re_λ	α_Π	Deviation vs. reference
isotropic1024coarse	≈ 433	0.9605 (ref.)	—
isotropic4096	≈ 610	—	1.39%
isotropic8192	≈ 1257	0.9584 ± 0.0501	0.22%

Table 1: Reynolds-independence of the shape parameter α_Π of the normalized spectral flux $\Pi(k)/\varepsilon$, over a factor ≈ 3 in Re_λ . isotropic8192 extraction: full uniform sampling of the domain (stride 128, oindex method), correcting a truncated-sampling artifact from an earlier attempt. Consistency check: Parseval ratio = 1.000.

Adopted formulation. The predicted value of α_Π remains con-

sistent with the reference to better than 5% over the tested range ($\text{Re}_\lambda \approx 433$ to 1257). No systematic drift with Reynolds number is observed.

Proposition 21.1 (Untested prediction). *$E(k) \propto \text{sech}^2(\alpha k)$ in the inertial range, a testable departure from Kolmogorov's $k^{-5/3}$ law; a Gaussian correction to the compensated spectrum is explicitly predicted (Version 2, Section 6.1).*

Part VII

The Representation Program, In Detail

22 Explicit Definition of the Projection Operators

22.1 The Three Known Projections

Definition 22.1 (Yang-Mills projection). $P_{\text{YM}}(\rho_0)(\mu^2) := \rho_0(\mu^2)$, the identity, giving the Källén-Lehmann density ρ_{KL} , related to the gluon propagator by $D(k^2) = \int_0^\infty \rho_{\text{KL}}(\mu^2)/(k^2 + \mu^2) d\mu^2$.

The width parameter α_{YM} must be extracted directly from $\rho_{\text{KL}}(\mu^2)$, not from $D(k^2)$; the two differ, since fitting a sigmoid to $D(k^2)$ gives the width of the Källén-Lehmann transform of ρ_0 , not of ρ_0 itself.

Definition 22.2 (Navier-Stokes projection). $P_{\text{NS}}(\rho_0)(k) := \int_0^k \rho_0(k') dk'$.

This is distinct from the measured flux $\Pi(k) = \int_0^k 2\nu k'^2 E(k') dk' \neq \int_0^k E(k') dk'$; the factor $2\nu k'^2$ comes from the stationary energy balance, not from P_{NS} as defined.

Definition 22.3 (Glass projection). $P_{\text{glass}}(\rho_0)(\lambda) := \rho_0(\sqrt{\lambda})/2\sqrt{\lambda}$,

giving the vibrational density of states $g(\omega)$ near the marginal point, $\lambda = \omega^2$.

22.2 What Was Actually Measured

The JHTDB validation (Part IV, Section 2) fit $\Pi(k)/\varepsilon$, not $P_{\text{NS}}(\rho_0)/P_{\text{NS}}(\rho_0)(\infty)$; the two differ by the dissipation weight $2\nu k'^2$. What is rigorously established: $\Pi(k)/\varepsilon$ is sigmoid-shaped in $\log k$ (qualitatively confirming the two-front prediction); its shape parameter is $\alpha_{\Pi} \approx 0.93$ – 0.96 depending on the snapshot set, stable in time at fixed Reynolds number. What is not established: the value of α_{NS} as defined for $P_{\text{NS}}(\rho_0) = \int_0^k E(k')dk'$, which requires fitting the sigmoid directly to $E(k)$ — the immediate next step, requiring only existing JHTDB data.

22.3 The Corrected Measurement Program

Open Question 22.4 (Cross-domain comparison protocol). *Compare, domain by domain: in Yang-Mills, fit $\rho_{\text{KL}}(\mu^2)$ (recovered from lattice $D(k^2)$ by Källén-Lehmann inversion) to a sigmoid in $\log \mu$, obtaining $\alpha_{\text{YM}}^{(\rho)}$; in Navier-Stokes, fit $E(k)$ directly to a sigmoid in $\log k$, obtaining $\alpha_{\text{NS}}^{(E)}$; in glasses, fit $g(\omega)/g(\omega_{\text{Debye}})$ near the marginal point, obtaining $\alpha_{\text{glass}}^{(g)}$. The cross-domain test is $\alpha_{\text{YM}}^{(\rho)} \stackrel{?}{=} \alpha_{\text{NS}}^{(E)} \stackrel{?}{=} \alpha_{\text{glass}}^{(g)}$.*

23 The Structural Invariant and Representability Tests

23.1 Why a Scalar Invariant Is the Wrong Object

An earlier conjecture, $I_i = \alpha_i \times \log(\omega_{i,\text{plateau}}/\omega_{c,i})$, was tested directly and rejected: in Navier-Stokes, fitting cumulative energy on JHTDB data gives $\alpha_{\text{NS}}^{(E)} \approx 0.44$; in Yang-Mills, recovering $\rho_{\text{KL}}(\mu^2)$ from lattice $D(k^2)$ by Källén-Lehmann inversion gives no stable sigmoid width across regularization schemes, varying by more than an order of magnitude. This instability is structural: ρ_{KL} in the confining phase is a sharply concentrated, sign-changing measure, while $E(k)$ is a smooth, monotonic cumulative flux.

23.2 The Correct Invariant: Structural, Not Scalar

What is established, independently of domain-specific data, is the result of Part III (Theorem 9.2, reformulated): under $H1$ – $H3$ and $H4'$, the spectral zero order is exactly $n = 2$, and no other order is structurally accessible *within this class*. The candidate cross-domain invariant is therefore not a scalar but a structural class: the type of singularity of ρ_0 at ω_c (double zero), the vanishing of the projective curvature $K = 0$, the value $\Theta = -1/2$, and the pure invariants J_1, J_2, J_3 of the kink itself.

Remark 23.1 (Precision added in Version 4). *Unlike Version 3’s presentation, this structural invariant is no longer presented as an automatic consequence of $H1$ – $H3$ alone: it explicitly requires $H4'$ (Part III). The projections P_i that export it to each physical domain therefore inherit this same conditional dependence.*

23.3 Representability as the Right Test

Navier-Stokes (established): sigmoid (hence $n = 2$ -compatible) shape of $\Pi(k)/\varepsilon$ confirmed on JHTDB data, RMS = 0.016, stable to 0.04% across snapshots, at three Reynolds points (Part IV, Section 2).

Yang-Mills (structural evidence, not yet a clean numerical test): the Oehme-Zimmermann positivity violation forces a sign change in ρ_{KL} , giving the qualitative marginal-crossing signature; the sigmoid fit to $D(k^2)$ itself gives RMS = 0.0016. What remains open: recovering ρ_{KL} robustly enough from lattice data to verify $n = 2$ directly.

Glasses (untested): no representability test has yet been performed.

24 The Theorem of Spectral Representation

Open Question 24.1 (Central open problem). *Let (M, g, Φ) satisfy: A1, M smooth with $g = 1/\rho$, $\rho > 0$; A2, $\Phi : M \rightarrow [-1, 1]$, $\Phi(-\infty) = -1$, $\Phi(+\infty) = 1$; A3, ρ_{net} (the projected net density, not the metric density ρ itself — clarification of Part IV, Section 1) has a double zero at some $\omega_c \in M$; A4, Φ satisfies the corrected KPP flow equation in the approach to the fixed point. Then: is every physical theory compatible with A1–A4 obtained by a morphism $P : M \rightarrow O$ in a class $\mathcal{P}$, unique up to isomorphism?*

If proved, Yang-Mills, Navier-Stokes, and glasses would no longer be chosen examples but necessary representations of (M, g, Φ) . Current status: conceptually largely open; the exact role of $H4'$ in axiom A3 (ensuring the double zero exists) is now explicit rather than implicit, which clarifies — without resolving — the problem.

Part VIII

Appendices

Appendix A: Verification Scripts

Scripts 1 through 9 are retained from Version 3 without modification (they verify unchanged results: kink uniqueness, Pöschl–Teller spectrum, non-circularity). Scripts 10 through 15 are new in Version 4.

Script 10: Sign Correction Verification

```
import sympy as sp

omega, D, lam, gam = sp.symbols('omega D lambda gamma', positive
    =True)
alpha = sp.sqrt(lam/(4*D))
rho_star = (sp.Rational(3,1)*lam/(2*gam)) * sp.sech(alpha*omega)
    **2

# Historical (INCORRECT) equation
residual_old = D*sp.diff(rho_star, omega, 2) + lam*rho_star -
    gam*rho_star**2
print("Residual (historical equation):", sp.simplify(
    residual_old))
# -> nonzero: 6*lambda**2*(cosh(...)-2)/(gamma*(cosh(...)+1)**2)
```

```

# Corrected equation
residual_new = D*sp.diff(rho_star, omega, 2) - lam*rho_star +
    gam*rho_star**2
print("Residual (corrected equation):", sp.simplify(residual_new
    ))
# -> 0, identically zero

```

Script 11: Genericity of the Simple Zero (Decoupled System)

```

import numpy as np
from scipy.optimize import brentq

def find_crossing_and_derivative(r):
    D, lam, gam = 1.0, 1.0, 1.0
    mu = r*lam
    alpha_p = np.sqrt(lam/(4*D)); alpha_m = np.sqrt(mu/(4*D))
    Ap0 = 3*lam/(2*gam); Am0 = 3*mu/(2*gam)
    def rho_net0(w):
        return Am0/np.cosh(alpha_m*w)**2 - Ap0/np.cosh(alpha_p*w)
        **2
    w_test = np.linspace(1e-6, 50, 5000)
    vals = np.array([rho_net0(w) for w in w_test])
    sc = np.where(vals[:-1]*vals[1:] < 0)[0]
    if len(sc) == 0: return None, None
    i = sc[0]
    w0 = brentq(rho_net0, w_test[i], w_test[i+1])
    h = 1e-6
    d1 = (rho_net0(w0+h)-rho_net0(w0-h))/(2*h)
    return w0, d1

```

```

for r in [1.001, 1.01, 1.1, 1.5, 2.0, 5.0, 10.0, 50.0, 100.0]:
    w0, d1 = find_crossing_and_derivative(r)
    print(f"r={r:8.3f} omega_0={w0:.4f} rho_net'(omega_0)={d1:.6f}")
# -> nonzero derivative across the entire tested range

```

Script 12: Coupled System Resolution (solve_bvp, No Ansatz)

```

import numpy as np
from scipy.integrate import solve_bvp
from scipy.optimize import brentq

D, lam, mu, gam = 1.0, 1.0, 1.5, 1.0
alpha_p0 = np.sqrt(lam/(4*D)); alpha_m0 = np.sqrt(mu/(4*D))
Ap0 = 3*lam/(2*gam); Am0 = 3*mu/(2*gam)
L = 20.0
w_mesh = np.linspace(-L, L, 1500)

rp_g = Ap0/np.cosh(alpha_p0*w_mesh)**2
drp_g = -2*alpha_p0*Ap0*np.tanh(alpha_p0*w_mesh)/np.cosh(alpha_p0*w_mesh)**2
rm_g = Am0/np.cosh(alpha_m0*w_mesh)**2
drm_g = -2*alpha_m0*Am0*np.tanh(alpha_m0*w_mesh)/np.cosh(alpha_m0*w_mesh)**2
y_guess = np.vstack([rp_g, drp_g, rm_g, drm_g])

def bc(ya, yb):
    return np.array([ya[0], yb[0], ya[2], yb[2]])

```

```

sol = None
for kappa in [0.0, 0.1, 0.2]:
    def ode(w, y, kappa=kappa):
        rp, drp, rm, drm = y
        d2rp = (lam*rp - gam*rp**2 + kappa*rp*rm) / D
        d2rm = (mu*rm - gam*rm**2 + kappa*rp*rm) / D
        return np.vstack([drp, d2rp, drm, d2rm])
    sol = solve_bvp(ode, bc, w_mesh, y_guess, max_nodes=100000,
        tol=1e-9)
    w_mesh, y_guess = sol.x, sol.y
    def rho_net(w):
        y = sol.sol(w); return y[0]-y[2]
    wc = np.linspace(0.05, 12, 200)
    vals = np.array([rho_net(w) for w in wc])
    sc = np.where(vals[:-1]*vals[1:]<0)[0]
    i = sc[0]
    w0 = brentq(rho_net, wc[i], wc[i+1])
    h = 1e-4
    d1 = (rho_net(w0+h)-rho_net(w0-h))/(2*h)
    print(f"kappa={kappa:.2f} omega_0={w0:.4f} rho_net'(omega_0)
        ={d1:.6f}")
# -> simple zero (n=1) for every tested kappa, /derivative/
    increasing with kappa

```

Script 13: Decomposition of the Double-Zero Mechanism

```

import sympy as sp

```

```

xi, D, lam, mu, Ap, Am, delta = sp.symbols(
    'xi D lambda mu A_p A_m delta', real=True)
alpha_p = sp.sqrt(lam/(4*D)); alpha_m = sp.sqrt(mu/(4*D))
Phi_m = sp.tanh(alpha_m*xi)
Phi_p = sp.tanh(alpha_p*(xi-delta))
rho_net = Am*(1-Phi_m**2) - Ap*(1-Phi_p**2)

series = sp.series(rho_net, xi, 0, 4).removeO()
c0 = sp.simplify(series.coeff(xi,0).subs(delta,0))
c1 = sp.simplify(series.coeff(xi,1).subs(delta,0))
c2 = sp.simplify(series.coeff(xi,2).subs(delta,0).subs(Ap,Am))
print("xi^0 (delta=0):", c0, "-> requires A_p=A_m")
print("xi^1 (delta=0):", c1, "-> automatically zero (parity)")
print("xi^2 (delta=0, A_p=A_m):", c2, "-> nonzero if lambda!=mu"
    )

```

Script 14: Overlap Integral $\mathcal{G}(\delta)$

```

import numpy as np
from scipy.integrate import quad

D, lam, mu, gam = 1.0, 1.0, 1.5, 1.0
alpha_p = np.sqrt(lam/(4*D)); alpha_m = np.sqrt(mu/(4*D))
Ap0 = 3*lam/(2*gam); Am0 = 3*mu/(2*gam)

def G(delta):
    f = lambda w: (Ap0/np.cosh(alpha_p*w)**2)*(Am0/np.cosh(
        alpha_m*(w-delta))**2)
    val, _ = quad(f, -50, 50, limit=200)
    return val

```

```

for d in np.linspace(-3, 3, 13):
    print(f"delta={d:6.2f} G={G(d):.6f}")
h = 0.05
d2G0 = (G(h)-2*G(0)+G(-h))/h**2
print(f"G''(0) = {d2G0:.4f} (negative => maximum at delta=0)")

```

Appendix B: Audit Chronology

The sign error was identified through cross-verification by three independent checks (direct symbolic computation, cross-proposed correction, confirmation against the June 2026 source). The status of $\Theta \rightarrow -1/2$ as a conditional corollary of $H4'$, rather than an independent consequence of the KPP system, was established through direct numerical resolution of the coupled system (without a kink ansatz), independently reproduced with agreement to the decimal place. Six attempts to derive $H4'$ were conducted and documented (Part III, Section 4). The pulse-interaction literature (Rademacher 2013, Ohta 2001) was identified as a relevant methodological framework but not directly applicable (absence of a diffusion-scale separation in RST).

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